4.9 Principal Value of Inverse Trigonometric Functions

Let us recall that the **principal value** of a inverse trigonometric function at a point x is the value of the inverse function at the point x, which lies in the range of principal branch. For instance, the principal value of $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$ is $\frac{\pi}{6}$, since $\frac{\pi}{6} \in [0, \pi]$. When there are two values, one is positive and the other is negative such that they are numerically equal, then the principal value of the inverse trigonometric function is the positive one. Now, we list out the **principal domain** and **range** of **trigonometric functions** and the **domain** and **range** of **inverse trigonometric functions**.

Function	Principal Domain	Range
sine	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$	$[-1,1]$
cosine	$[0, \pi]$	$[-1,1]$
tangent	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	$\mathbb{R}$
cosecant	$\left[\frac{-\pi}{2}, \frac{\pi}{2}\right] \setminus \{0\}$	$\mathbb{R} \setminus (-1, 1)$
secant	$[0, \pi] \setminus \left\{\frac{\pi}{2}\right\}$	$\mathbb{R} \setminus (-1,1)$
cotangent	$(0, \pi)$	$\mathbb{R}$

Inverse Function	Domain	Range of Principal value branch
$\sin^{-1}$	$[-1,1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1}$	$[-1,1]$	$[0, \pi]$
$\tan^{-1}$	$\mathbb{R}$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
cosec^{-1}	$\mathbb{R} \setminus (-1,1)$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \setminus \{0\}$
$\sec^{-1}$	$\mathbb{R} \setminus (-1,1)$	$[0, \pi] \setminus \left\{\frac{\pi}{2}\right\}$
$\cot^{-1}$	$\mathbb{R}$	$(0, \pi)$

Example 4.12

Find the principal value of

(i) $\text{cosec}^{-1}(-1)$ (ii) $\sec^{-1}(-2)$.

Solution

(i) Let $\text{cosec}^{-1}(-1) = y$. Then, $\text{cosec}\, y = -1$

Since the **range of principal value branch** of $y = \text{cosec}^{-1} x$ is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \setminus \{0\}$ and

$\text{cosec}\left(-\frac{\pi}{2}\right) = -1$, we have $y = -\frac{\pi}{2}$. Note that $-\frac{\pi}{2} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \setminus \{0\}$.

Thus, the principal value of $\text{cosec}^{-1}(-1)$ is $-\frac{\pi}{2}$.

(ii) Let $y = \sec^{-1}(-2)$. Then, $\sec y = -2$.

By definition, the **range of the principal value branch** of $y = \sec^{-1} x$ is $[0, \pi] \setminus \left\{\frac{\pi}{2}\right\}$.

Let us find y in $[0, \pi] - \left\{\frac{\pi}{2}\right\}$ such that $\sec y = -2$.

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